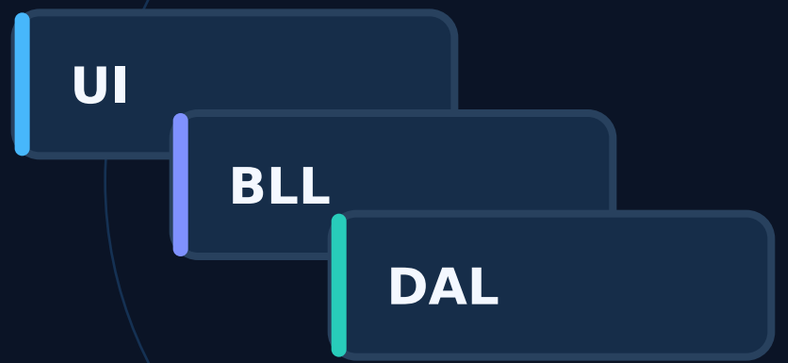
TECHNICAL APPENDIX / C

SYSTEM ARCHITECTURE

& DATABASE DESIGN

DVLD / Driving License Management System

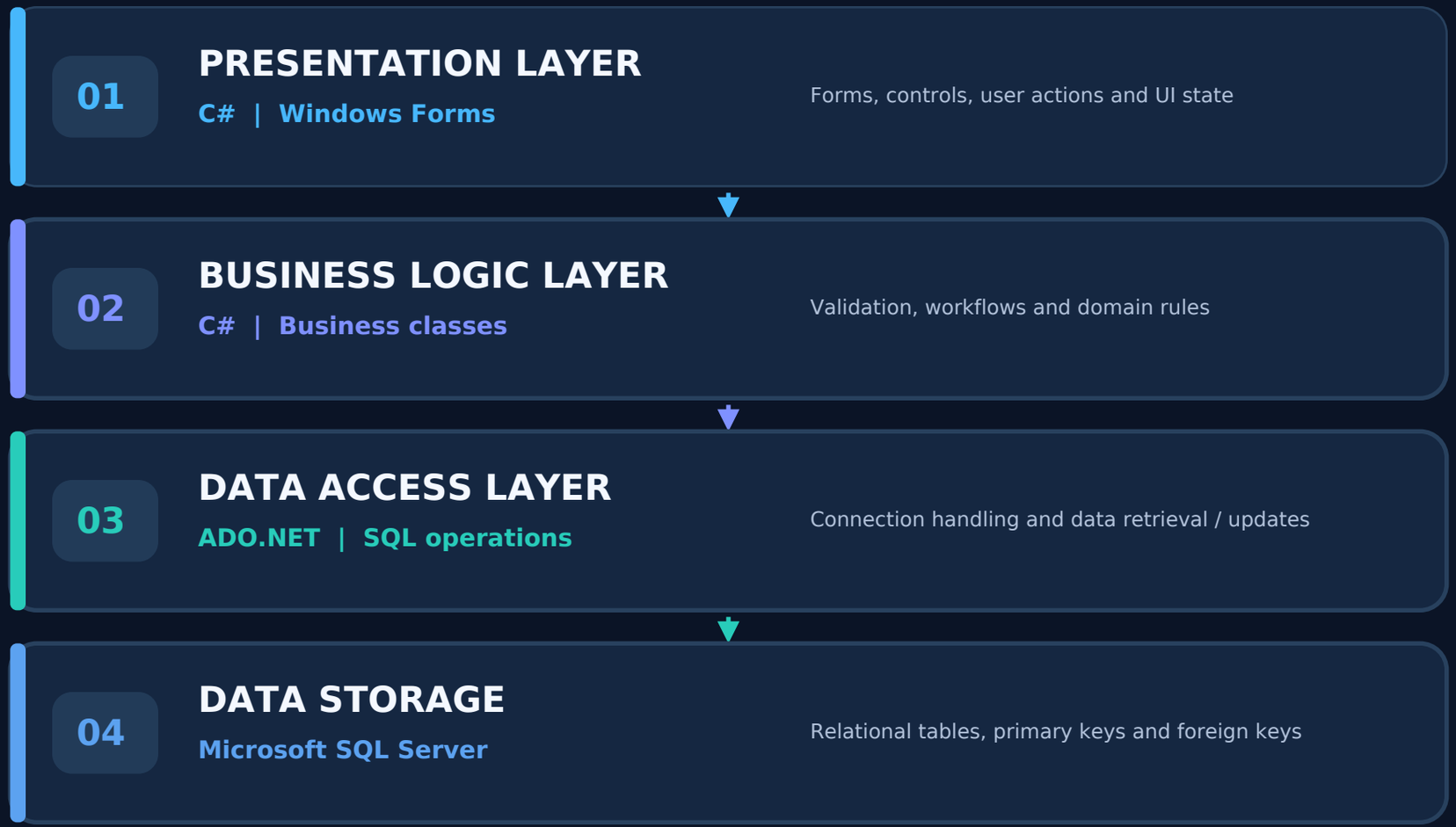
Three-tier architecture + relational database ERD



01 / SYSTEM ARCHITECTURE

Three-Tier Application Architecture

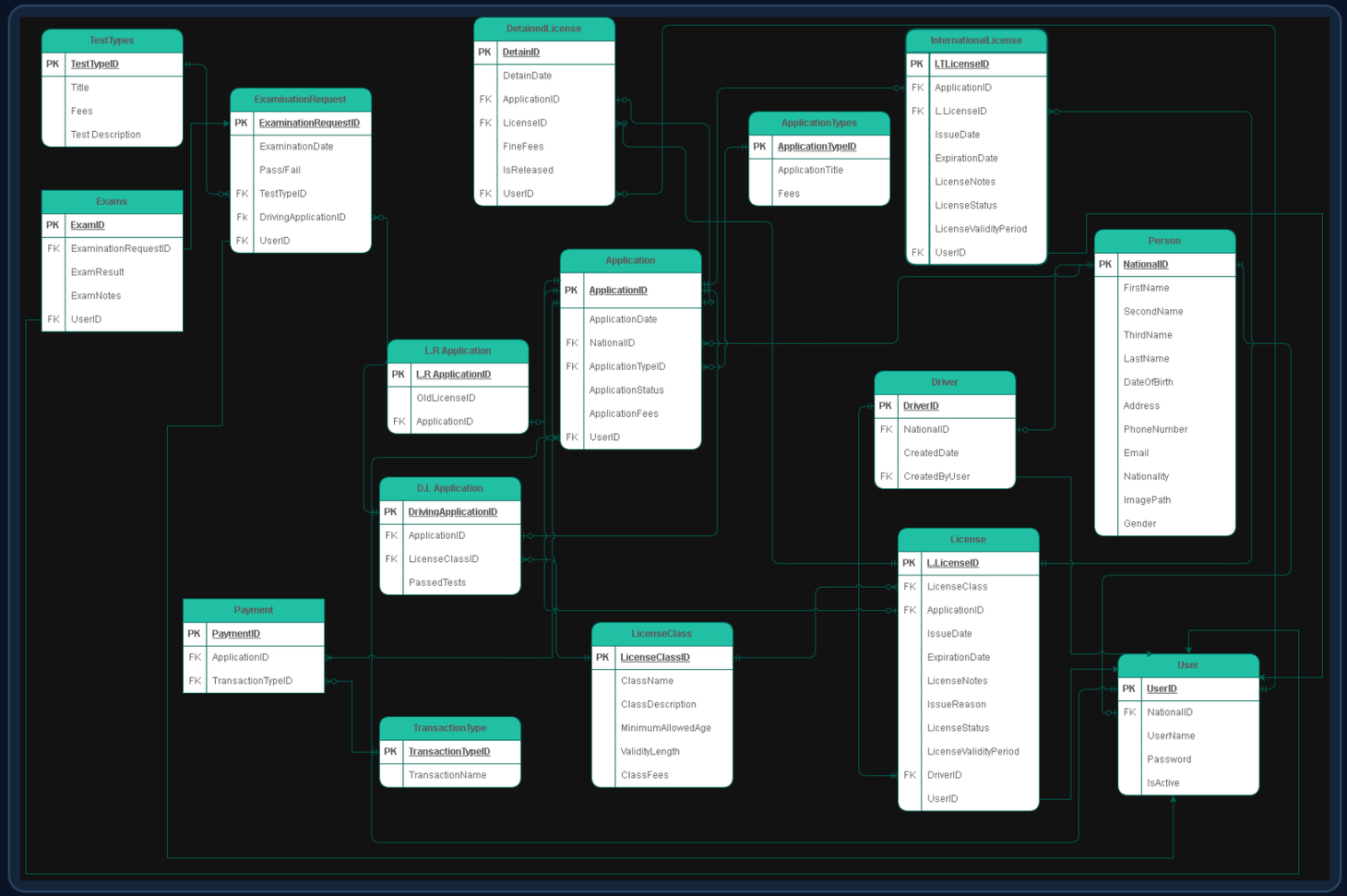
Layered separation of interface, business rules, and persistence.



02 / DATABASE ENTITY-RELATIONSHIP DIAGRAM

Relational Data Model

Source diagram supplied with the project. Original entity and relationship details preserved.



03 / DESIGN OVERVIEW

Architecture & Data Design Notes

Key design decisions represented by the architecture and database ERD.

01 Separation of concerns

UI manages interaction; business classes implement workflows; DAL manages SQL operations.

02 Controlled data access

ADO.NET mediates access to the SQL Server database rather than placing SQL in UI forms.

03 Relational integrity

Entity identifiers and foreign-key references connect people, drivers, users, applications and licenses.

04 Domain workflows

The schema supports licensing applications, examinations, payments and license-related operations.

Implementation stack: C# WinForms / C# BLL / ADO.NET / SQL Server